

Assessing the Quality of DESI Spectroscopic Survey

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Introduction

Automating the detection of anomalous DESI spectra is essential for high-throughput cosmological surveys. By integrating UMAP with Friends-of-Friends clustering, our pipeline extends [3] and replaces manual inspection (see e.g., [1]), enabling fast and reproducible identification of rare or corrupted spectra.

What is DESI?

The **Dark Energy Spectroscopic Instrument** (DESI) is mapping the 3D distribution of 35 million galaxies and quasars up to $z \approx 1.6$.

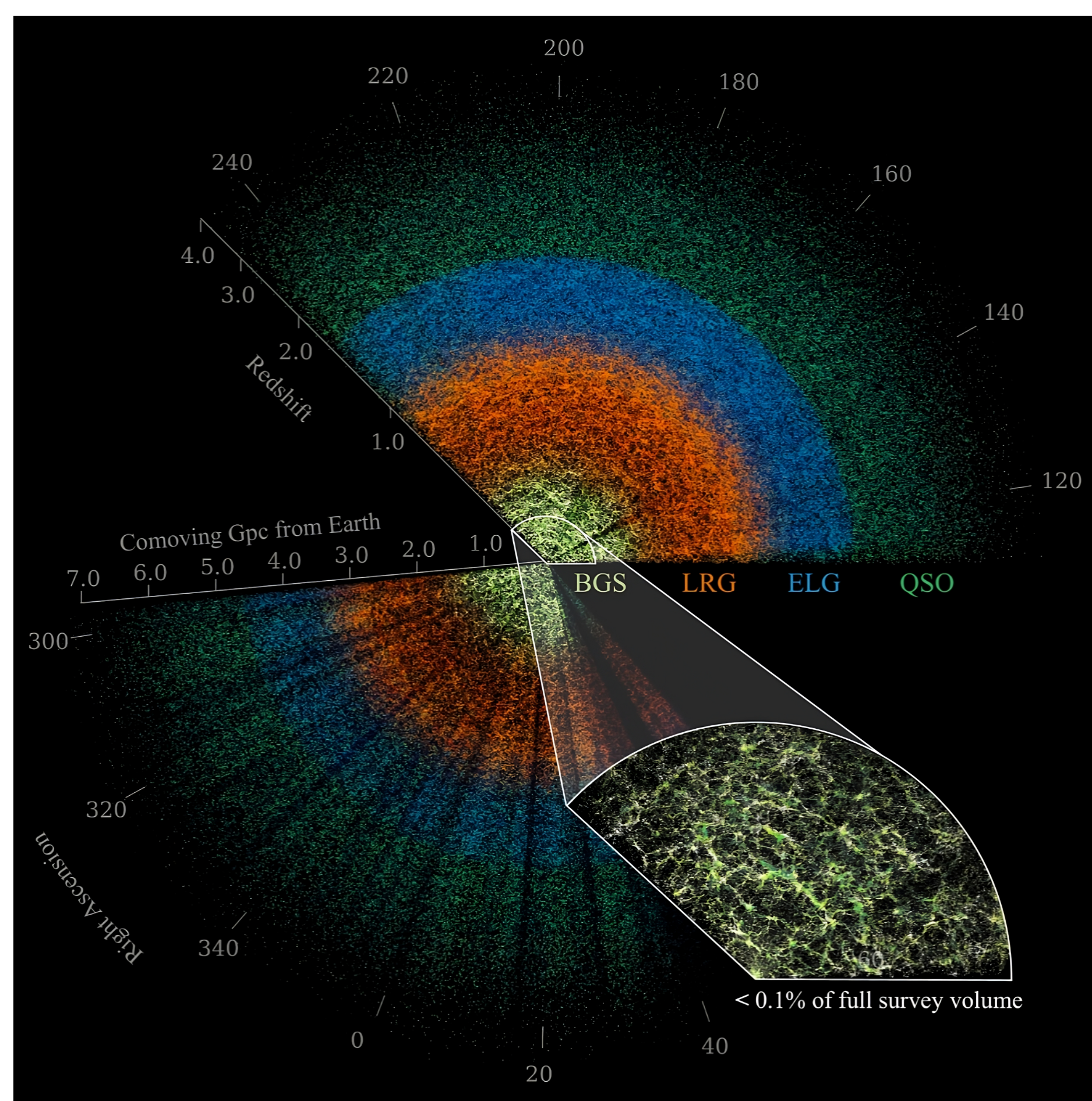


Figure 1: Each dot represents a galaxy or quasar, colors indicate redshift bins.

Reliable, automated quality checks are crucial to ensure data integrity at this scale [2].

Data

Each circular tile contains 5 000 fibers collecting coadded spectra in three bands: **B**, **R**, and **Z**.

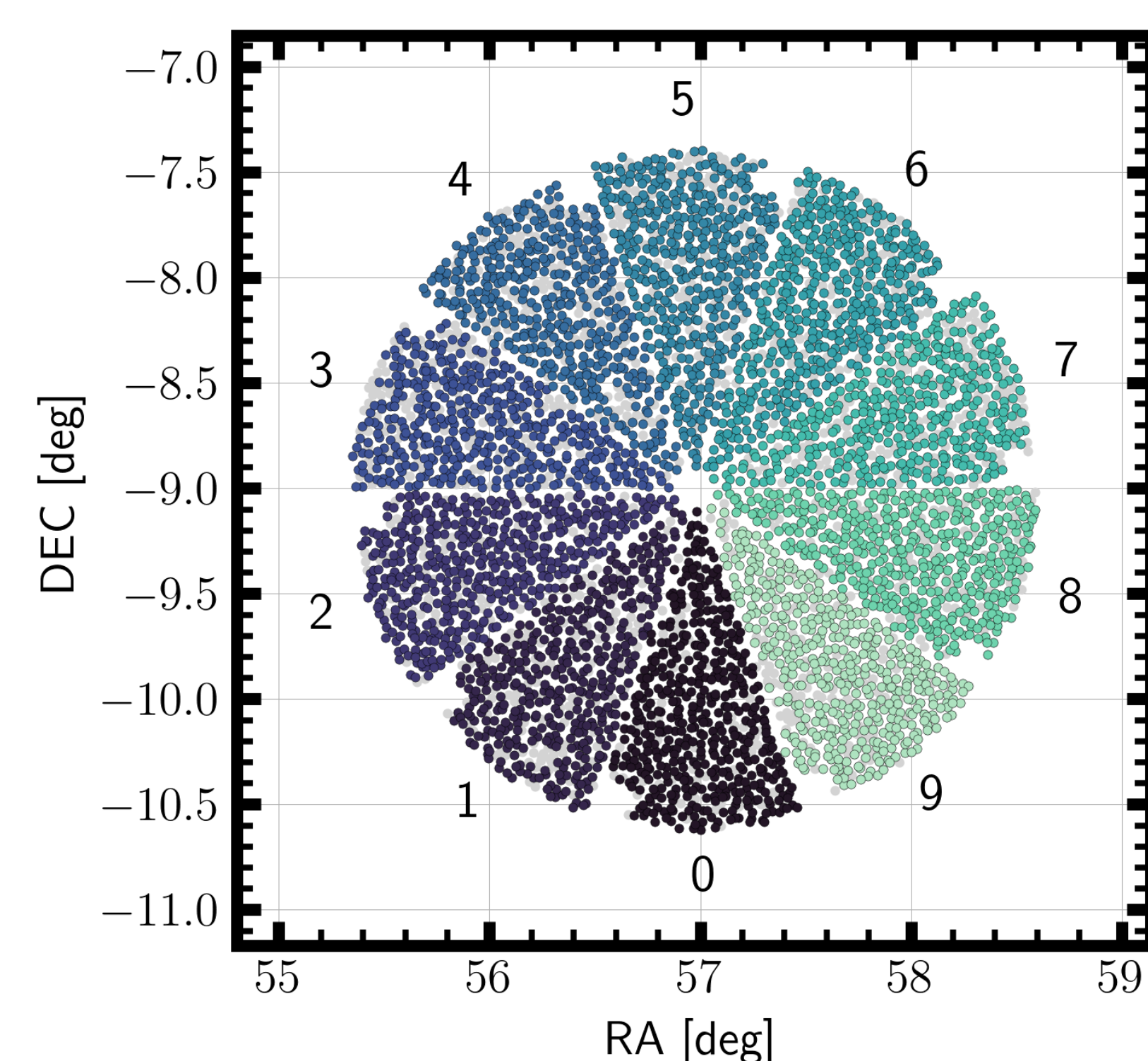


Figure 2: Tile layout: fibers 0–4999 grouped by petal number.

We focus on LRG, ELG, QSO targets, plus standard stars.

UMAP & FoF

UMAP embeds spectra into 2D so similar spectra cluster together. **FoF** then builds a graph on this embedding and links neighbors and finds outliers.

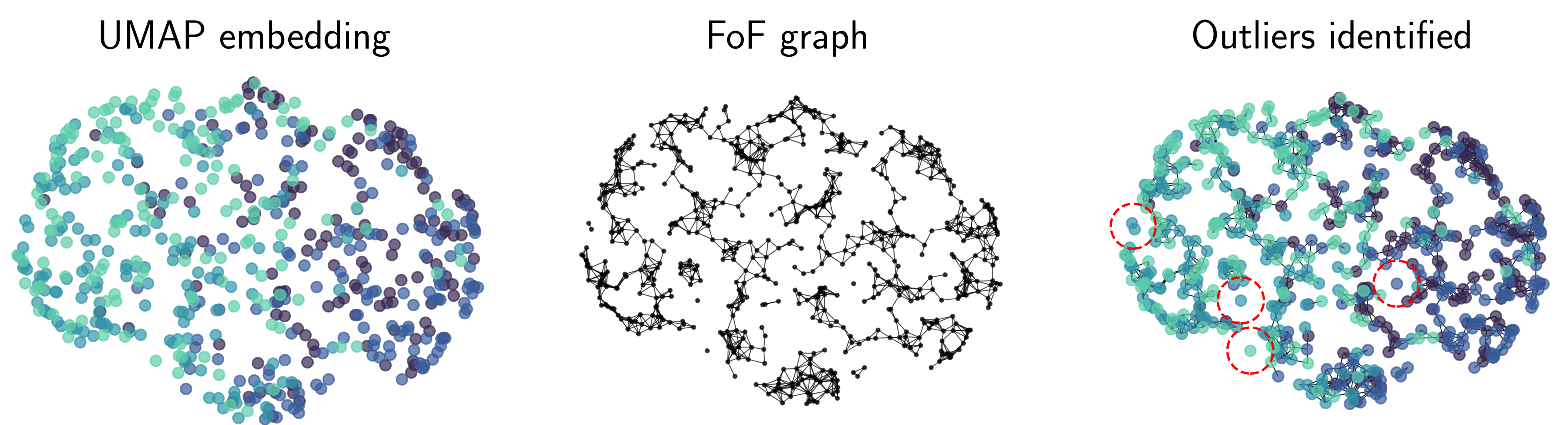


Figure 3: Left: 2D embedding, each point = one spectrum. Center: FoF graph edges link neighboring spectra. Right: Red circles mark outliers.

Together, they provide an efficient, fully unsupervised quality check by highlighting spectra that deviate from the main populations.

Analyze Spectrum

This spectrum at $z = 0.672$ stands out as an outlier in the pipeline, probably due to its unusually broad emission around 9000 Å and sharp features beyond 9500 Å. No neighboring fibers overlap this position, ruling out instrumental artifacts.

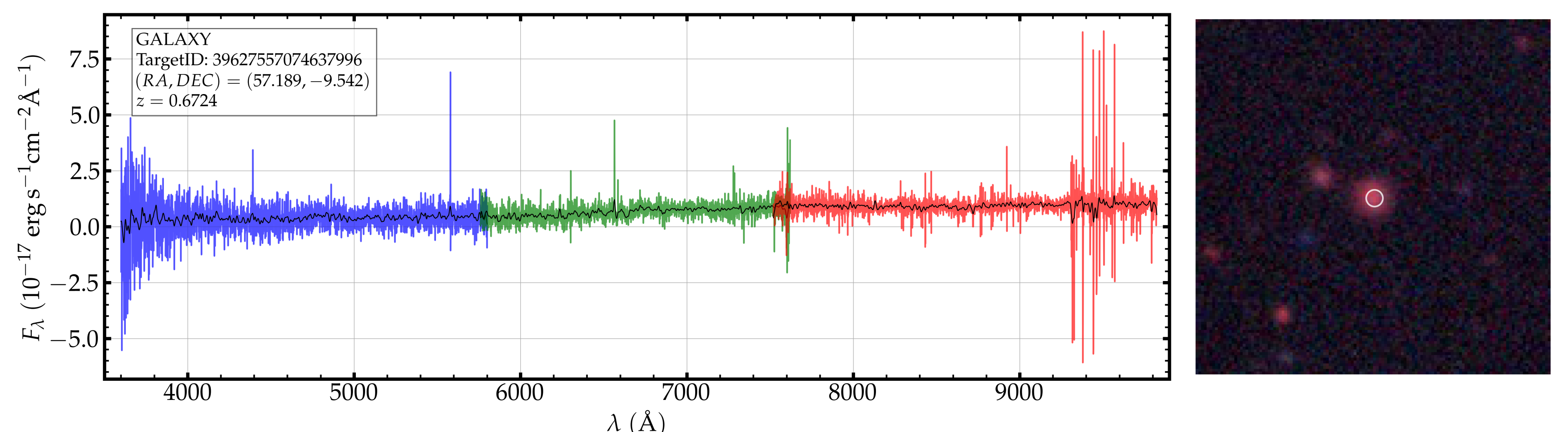


Figure 4: Left: Flux in b, r, and z bands. Right: Sky image at fiber location, red circle = target.

Conclusions & Next Steps

- End-to-end UMAP+FoF pipeline processes ~ 4000 spectra per tile in under 60 s (10.3 ms/spec). Performance scales linearly:

$$t(N) = \left(10.33 \frac{\text{s}}{10^3 \text{ spec}} \right) N \quad (1)$$

- Fully automated, reproducible pipeline suitable for real-time monitoring.
- Next steps include deploying on DESI data server for nightly quality assurance and integrate feedback for parameter tuning.

References

- [1] T.-W. Lan et al. “The DESI Survey Validation: Results from Visual Inspection of Bright Galaxies, Luminous Red Galaxies, and Emission-line Galaxies”. In: *The Astrophysical Journal* 943.1 (Jan. 2023), p. 68. ISSN: 1538-4357. DOI: 10.3847/1538-4357/aca5fa. URL: <http://dx.doi.org/10.3847/1538-4357/aca5fa>.
- [2] A. D. Myers et al. “The Target-selection Pipeline for the Dark Energy Spectroscopic Instrument”. In: *The Astronomical Journal* 165.2 (Jan. 2023), p. 50. ISSN: 1538-3881. DOI: 10.3847/1538-3881/aca5f9. URL: <http://dx.doi.org/10.3847/1538-3881/aca5f9>.
- [3] J. F. Suárez Pérez. “Artificial Intelligence in astronomy: machine learning and deep learning approaches to DESI data”. PhD thesis. Universidad de los Andes, 2023. URL: <http://hdl.handle.net/1992/68996>.